Investigating Tensor Network Models Based on Matrix Product States

Summer Project Report 2021

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Abstract

Using tensor network techniques, we compute the magnetization and spin-spin correlation for the Affleck-Kennedy-Lieb-Tasaki (AKLT) Hamiltonian in 1D spatial dimension. Tensor Network methods have been used to study physical observables of various systems in the thermodynamic limit. We study the AKLT model using density matrix renormalization group (DMRG) algorithm and the lowest energy states as matrix product state (MPS) wavefunction. We show our results of magnetization and spin-spin correlation agree with the exact values.

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Pokhee Saharia
Guwahati, July 3, 2021

1 Introduction

The study of quantum many-body systems prove crucial towards the understanding of several problems in condensed matter physics. The most common way of approaching these problems involve the reduction of the models to the simplest forms, which can be solved easily through numerical methods, while representing the relevant interactions.

1.1 Hilbert Spaces and the Area Law

Tensor network (TN) methods are one such class of numerical simulation algorithms to solve quantum systems, where the wave function of the system is modeled by a network of interconnected tensors.[2] Working through visual diagrams is at the core of TN algorithms, thus making them different from other numerical methods to study quantum systems, which usually specify coefficients of the wave functions. Other methods lead to the problem of a concerned Hilbert space that is far too large. However, not all quantum states in the Hilbert space of a many-body system are equal - the relevant states correspond to Hamiltonians whose interactions between the particles tend to be local. It can be proven that low-energy eigenstates of gapped hamiltonians with local interactions obey the area law, which means that the entanglement entropy of a region of space tends to scale, for large enough regions, as the size of the boundary of the region and not as the volume. Therefore, it follows naturally that not any state in the Hilbert space can be a low energy state of a gapped, local Hamiltonian. Only those states which follow the area law are relevant to us. And the family of TN states is the one which targets this corner of relevant states.

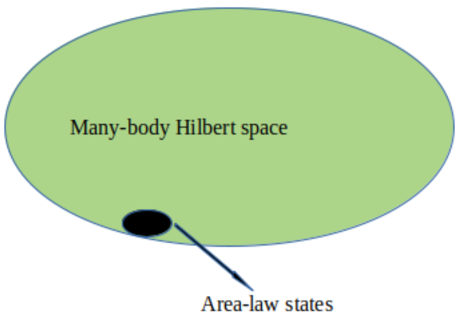

Figure 1: Diagrammatic representation of the area-law states in the many-body Hilbert space.

1.2 Tensor Network Theory

Tensors are basically multidimensional arrays of complex numbers. The dimension of a tensor is referred to as its ‘rank’; which makes scalars rank 0 tensors, vectors rank 1 tensors, matrices rank 2 tensors and so on.

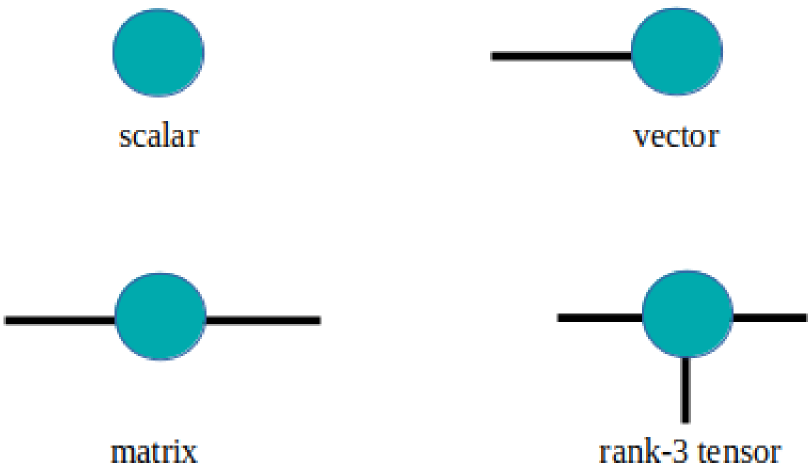

Figure 2: Diagrammatic representation of tensors of different ranks.

A very common operation performed on tensors is the index contraction, or referred simply to as, contraction. It is the sum over all possible values of the repeated indices of a set of tensors. Consider the contraction

$$C_{\alpha\gamma} = \sum_{\beta=1}^{D} A_{\alpha\beta} B_{\beta\gamma}. \tag{1}$$

Figure 3: Diagrammatic representation of tensor network for Eq. (1).

In Eq. (1), the tensors A and B are being contracted over the index β, resulting in the tensor C. The index β can take D different values in this example. In Fig. 3 we show the diagrammatic representation of this tensor contraction.

Tensor networks, are, essentially a group of tensors whose indices have been contracted according to some pattern. When all indices are contracted, the result is a scalar; when some are contracted it results in another tensor with a certain number of open indices.

As mentioned before, tensor network theory is largely a visual one, involving diagrammatic representations of tensor contraction - referred to as *tensor network diagrams.*

Consider the contraction

$$F_{\gamma\omega\rho\sigma} = \sum_{\alpha,\beta,\delta,\nu,\mu=1}^{D} A_{\alpha\beta\delta\sigma} B_{\beta\gamma\mu} C_{\delta\nu\mu\omega} E_{\nu\rho\alpha}. \tag{2}$$

In Fig. 4 we show the tensor network diagram representing this contraction.

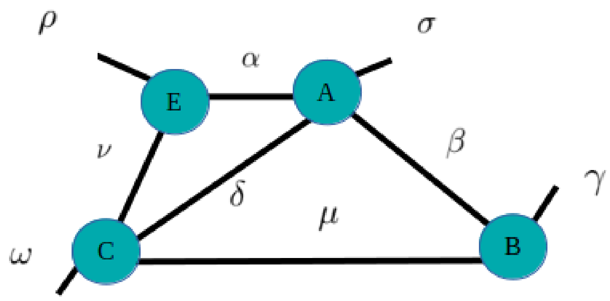

Figure 4: Diagrammatic representation of tensor network for Eq. (2).

Sometimes, the same result for a tensor contraction can be achieved through different number of operations. Therefore, it is necessary to find the optimal order in which the indices are to be contracted such that the computational cost is kept to a minimum.

Our discussion in this report being limited to one dimensional systems, we will focus on the quantum system that is most easily modeled by TN algorithms - the *Heisenberg Spin Chain.*

2 The Affleck-Kennedy-Lieb-Tasaki (AKLT) model

The one dimensional Heisenberg spin chain consists of N sites, on each of which sits a spin-1/2 particle and generates a two dimensional Hilbert space (since they are, at any point, in a linear combination of a spin up and spin down state).

The AKLT model is a spin-1 chain in which the spins interact via the Heisenberg interaction.[4] This model, which one may essentially term as the Heisenberg spin-1 chain, is a slight deviation from the original spin-1/2 model. Here, we visualize each spin-1 particle on the sites as consisting of two virtual spin-1/2 particles. These virtual particles form a spin singlet (a valence bond) with the virtual particles on their neighbouring sites. The tensor diagram is shown in Fig. 5.

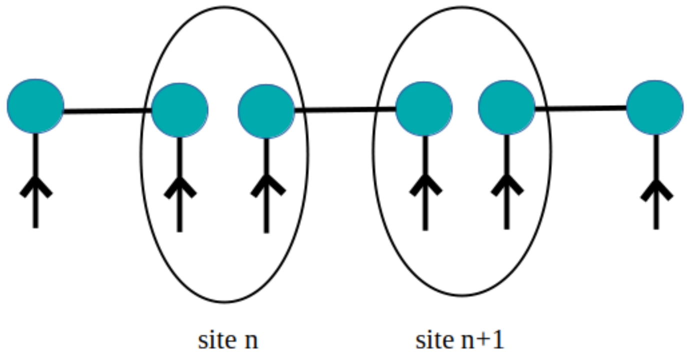

Figure 5: Diagrammatic representation of singlet formation in the spin-1 chain. Each of the circles represent spin-1/2 tensors.

Figure 6 shows the mapping of the Hilbert space of one spin-1 into a subspace of the product space of two spin-1/2s. The triangles are diagrammatic representations of the isometries, they do not represent tensors, unlike the circles in the diagram.

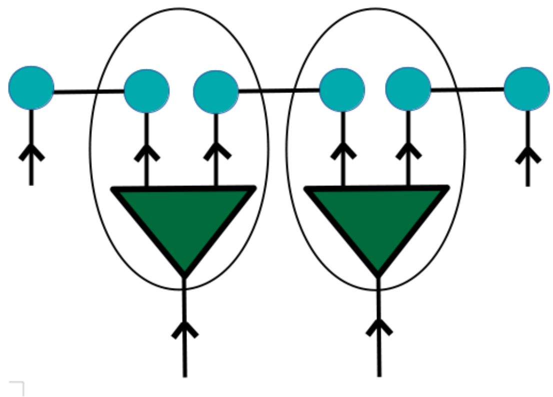

Figure 6: Diagrammatic representation of the mapping of the spin-1 Hilbert space.

The tensors marked inside the same oval are contracted, which gives the rank-3 tensors that constitute the AKLT state as an MPS, as shown in Fig. 7.

The ground state of the AKLT model can be represented as a valence bond solid with a single valence bond connecting every neighboring pair of sites, thus making it a very efficient model for studying more complicated spin systems. In practical terms, the contraction of tensors correspond to valence bonds in the model. The AKLT model, unlike its precursor and motivation, the Majumder-Ghosh model (Addendum), is not a frustrated chain; leading to the fact that its ground states minimize all local interactions simultaneously.

The ground state of the AKLT model is exactly solvable and is known to be a *matrix product state*, which is a class of tensor networks.

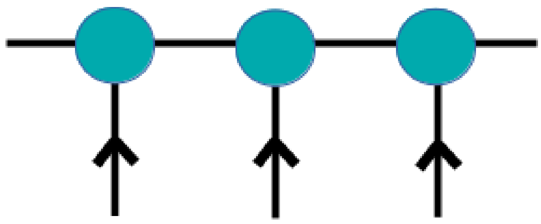

Figure 7: The spin-1 Heisenberg chain, with the AKLT ground state.

2.1 Matrix Product States (MPS)

Matrix product states, commonly abbreviated as MPS, are a family of tensor-network states that correspond to a one-dimensional array of tensors.[2] The bond indices that connect the tensors can take up to D values (the bond dimension of the tensor network). The open indices correspond to the physical degrees of freedom of the local Hilbert space. MPS can have either open or periodic boundary conditions.

(a) MPS with open boundary conditions. (b) MPS with periodic boundary conditions.

Figure 8: Types of matrix product states.

The mathematical formulation of a quantum state which is an MPS is as follows

$$|\psi\rangle = \sum_{\{s\}} \mathrm{Tr}\left[{A_1}^{(s_1)}{A_2}^{(s_2)}\cdots{A_N}^{(s_N)}\right]|s_1 s_2 \cdots s_N\rangle \tag{3}$$

where, ${A_i}^{(s_i)}$ are complex square matrices of a certain local dimension.

For the AKLT model

$$
\begin{aligned}
A^{+} &= \sqrt{\frac{2}{3}}\sigma^{+} = \begin{bmatrix} 0 & \sqrt{\frac{2}{3}} \\ 0 & 0 \end{bmatrix}. \\
A^{0} &= \frac{-1}{\sqrt{3}}\sigma^{z} = \begin{bmatrix} \frac{-1}{\sqrt{3}} & 0 \\ 0 & \frac{1}{\sqrt{3}} \end{bmatrix}. \\
A^{-} &= -\sqrt{\frac{2}{3}}\sigma^{-} = \begin{bmatrix} 0 & 0 \\ -\sqrt{\frac{2}{3}} & 0 \end{bmatrix},
\end{aligned}
$$

where the σ's are the Pauli matrices.

Visualization of an MPS wave function as a tensor network diagram would look like Fig. 9.

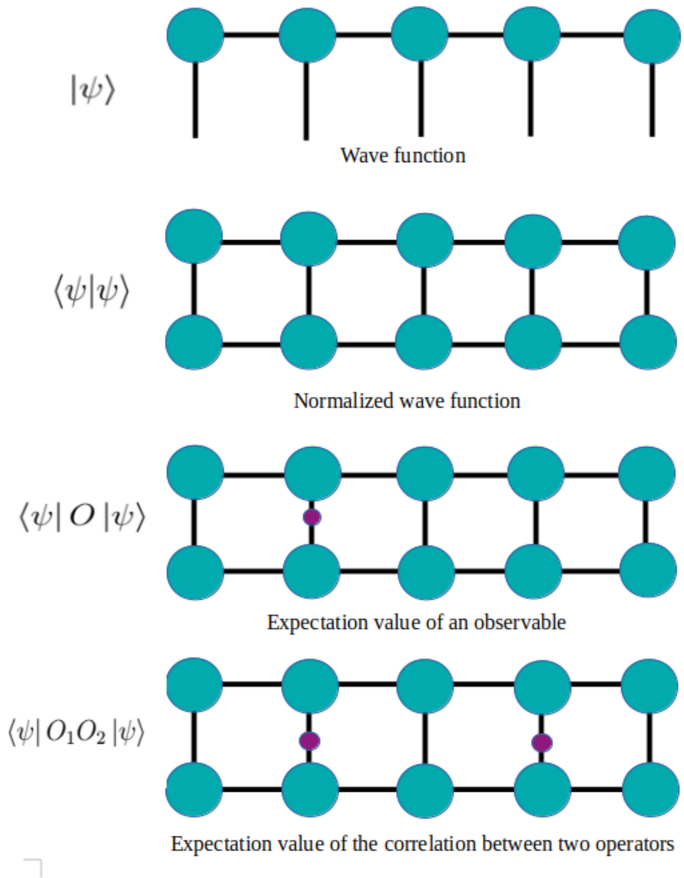

Figure 9: Diagrammatic representation of an MPS wave function.

2.2 The AKLT Hamiltonian Matrix Product Opertaor (MPO)

The Hamiltonian of the AKLT state is given by the expression

$$\hat{H} = \sum_{i} \left(\vec{S}^{[j]} \vec{S}^{[j+1]} + \frac{1}{3} \left(\vec{S}^{[j]} \vec{S}^{[j+1]} \right)^2 \right), \tag{4}$$

where $\vec{S}^{[i]}$ is the vector of spin-1 operators at site j. The Hamiltonian can be expanded and expressed in terms of operators S^z, S^+ and S^- at each site in the following form

$$\begin{aligned} \hat{H} \;=\;& {S^z}_j {S^z}_{j+1} + \frac{1}{2} {S^+}_j {S^-}_{j+1} + \frac{1}{2} {S^-}_j {S^+}_{j+1} \\ &+ \frac{1}{3} {S^z}_j {S^z}_j {S^z}_{j+1} {S^z}_{j+1} + \frac{1}{3} {S^z}_j {S^+}_j {S^z}_{j+1} {S^-}_{j+1} \\ &+ \frac{1}{3} {S^z}_j {S^-}_j {S^z}_{j+1} {S^+}_{j+1} + \frac{1}{12} {S^+}_j {S^+}_j {S^-}_{j+1} {S^-}_{j+1} \\ &+ \frac{1}{12} {S^-}_j {S^-}_j {S^+}_{j+1} {S^+}_{j+1} + \frac{1}{6} {S^-}_j {S^+}_j {S^+}_{j+1} {S^+}_{j+1}, \end{aligned} \tag{5}$$

where, S_z is the z component of the spin operator; S^+ and S^- are the spin raising and lowering operators, respectively. They are given by

$$S^{\pm} = S^x \pm i S^y.$$

N spin-1 sites are constructed at first and the Hamiltonian is inserted by a site-by-site decomposition; by following a standard *ITensor* syntax.

3 Density Matrix Renormalization Group (DMRG)

Most quantum many body systems are complex to deal with. Exact deductions of their physical quantities are rarely possible. Such situations compel us to turn to variational methods to extract properties of a system at low-energy limits. DMRG is one such method, and we use it here for our model of the Heisenberg spin chain.

The basic algorithm behind the DMRG technique is as follows

- In the initial step, left and right blocks, say A and B, are introduced, which contain one spin site each. This system is easily solved by diagonalizing the Hamiltonian.
- In subsequent steps, pairs of spins are inserted between the blocks, and the total spins present are again divided into two blocks. This increases the length of our

spin chain by 2 in each step, and length of each block by 1.

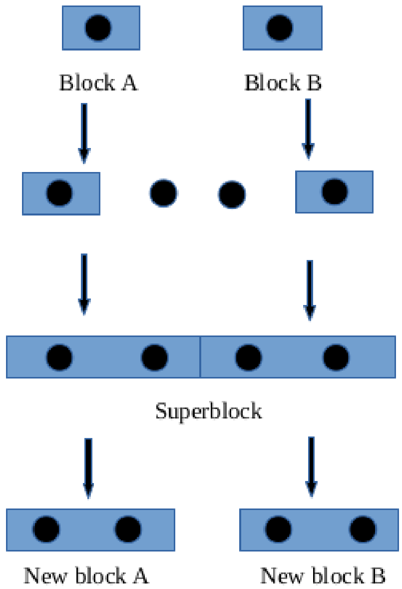

Figure 10: Diagrammatic representation of the initial DMRG step.

Any state in the intermediate superblock chain can be expressed as

$$
\begin{aligned}
|\psi\rangle &= \sum_{a_A\sigma_A\sigma_B a_B} \psi_{a_A\sigma_A\sigma_B a_B} |a\rangle_A |\sigma\rangle_A |\sigma\rangle_B |a\rangle_B \\
&= \sum_{i_A, j_B} \psi_{i_A j_B} |i\rangle_A |j\rangle_B .
\end{aligned} \tag{6}
$$

Here, $\{|\sigma\rangle_A\}$ and $\{|\sigma\rangle_B\}$ are the basis states with dimension d of the site next to A and B, respectively. $\{|i\rangle_A\}$ and $\{|i\rangle_B\}$ are the combined states for the new blocks A and B with dimension Dd.

Therefore, the ground state will be given by

$$E = \langle\psi_0| \hat{H}_{A..B} |\psi_0\rangle . \tag{7}$$

- Therefore, the Hilbert space spanned in each step grows exponentially, by d^l, where d is the dimension of the spin at one site and l is the length of the chain at a particular step.

- A truncation limit D is provided to set the reduced block space dimension. Hence, for Block A, at any step, we have the D-dimension basis states as $\{|a_l\rangle_A\}$, where l is the length of the spin chain at that step.

3.1 DMRG in Context of Tensor Networks

The entire computational process in this report has made extensive use of the ITensor library in C++.[3]

The DMRG algorithm in the ITensor library is basically iterative Singular Value Decomposition (SVD) procedures run on consecutive tensors. The visualization of a tensor network from a spin chain is discussed in a previous section.

The code works with Hamiltonians in matrix product operator (MPO) form.

The *dmrg* function accepts the following arguments:

- The Hamiltonian H, in an MPO form.
- The randomly initialized MPS M, formed using (sites) as an argument which refers to the construction of a spin chain with a certain number of sites.
- Sweeps, referring to the number of sweeps the code will make on the system
- "Quiet," which suppresses most output information about the computation except a short summary of each DMRG sweep.

The function returns the ground state energy of the system.

4 Measuring the Properties of the AKLT State

4.1 Measurement of the Magnetization at Each Site

Measurement of the magnetization involves the measurement of the expectation value of the S^z operator at each site. The algorithm to carry out this calculation at a particular site is as follows

- The tensors to the left and the right of the site are gauged to be orthogonal first.
- The MPS site tensor is extracted, for the bra-ket operation
- The tensor extracted in the above step is converted into a row vector, and its entries conjugated. We now have the bra.
- The S_z operator at the site is also retrieved.

- The expectation value is measured by contracting the bra, the operator and the ket.

The above steps are repeated through a for loop for every site, and the results for $N = 10, 20, 30$. They are shown in Fig. 11.

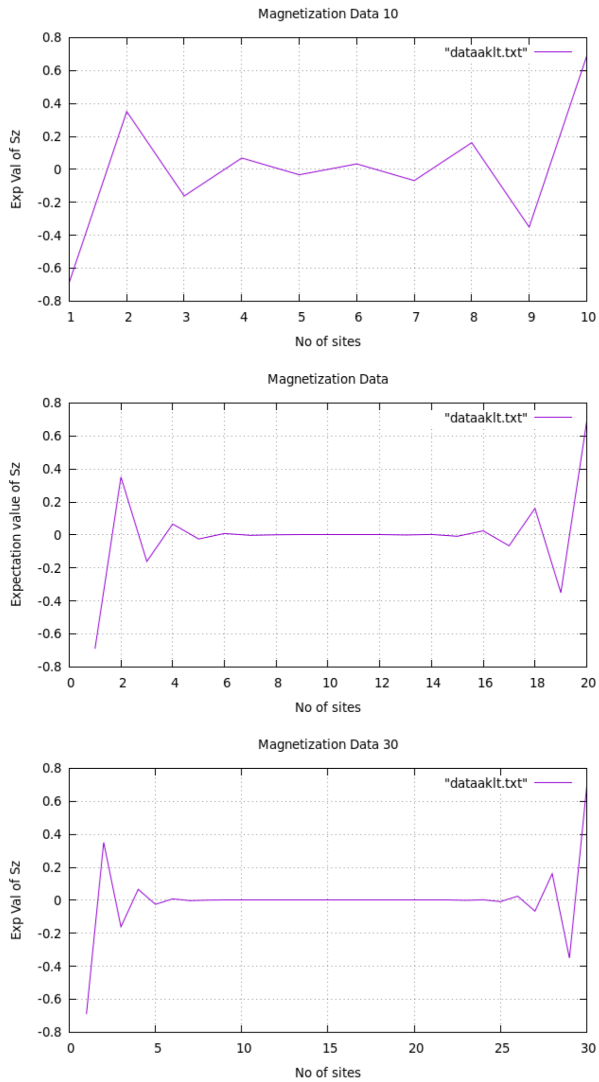

Figure 11: Magnetization at each site for the AKLT model.

The computed data was compared with the analytical data for $N = 50$ sites, which is given by the expression

$$S^z = \frac{\left(\frac{-1}{3}\right)^j - \left(\frac{-1}{3}\right)^{N-j+1}}{\frac{1}{2}\left[1 + \frac{-1}{3}^N\right]}. \tag{8}$$

From Fig. 13 we see that the computed and analytical results are in perfect agreement.

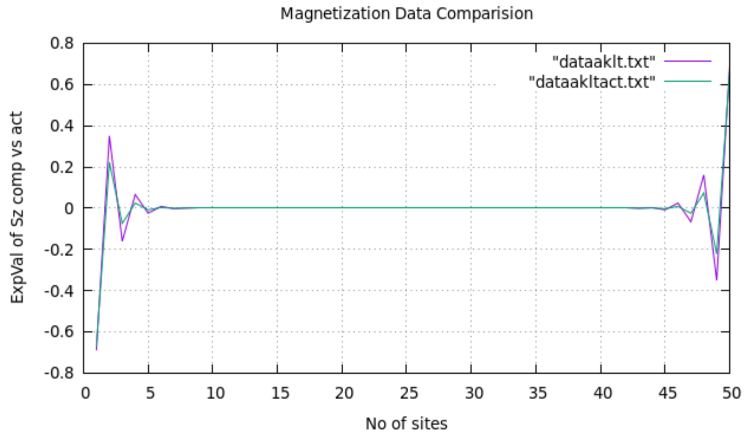

Figure 12: Magnetization at each site for the AKLT model. Here, $N = 50$. The numerical result is compared with the analytical result.

4.2 Spin-spin correlation ($\langle S^z S^z \rangle$)

The spin-spin correlation of the model is calculated in a similar manner to the magnetization, except that we have to extract operators associated with two sites.

The algorithm to calculate the correlator data of two consecutive sites, say j and $j+1$, is as follows

- As before, we gauge the tensors to the left and right of the concerned site, j, to be orthogonal.
- The ket vector is extracted by contracting the two MPS site tensors, at site j and $j+1$. It is converted into a row vector and the values conjugated to give the bra.
- The S^z operators are retrieved at sites j and $j+1$, contracted together, and contracted separately again.

The above steps are repeated for each site in the chain.

The computed results are plotted against the analytical ones given by the expression

$$y = \frac{\left(\frac{-2}{9}\right) + 2\left(\frac{-1}{3}\right)^N}{\left(\frac{1}{2}\right)\left(1 - \left(\frac{-1}{3}\right)^N\right)}. \tag{9}$$

It is seen that the computed and analytical results are in perfect agreement, except at the boundaries. This can be attributed to the fact that the boundary conditions while

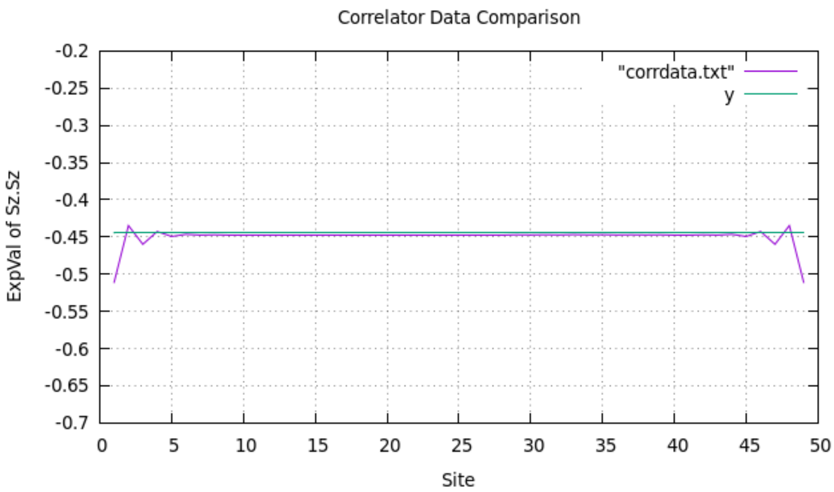

Figure 13: Spin-spin correlation for the AKLT model. Here, $N = 50$. The numerical result is compared with the analytical result.

computing the data (which were open), were not the same as the boundary conditions considered while deriving the analytical result.

4.3 Spin-spin correlation ($\langle S^+S^- \rangle$) / ($\langle S^-S^+ \rangle$)

In the same way as above, the spin-spin correlation for the S^+ and S^- operators is computed. Instead of extracting the S^z operator, S^+ at site j and S^- at site $j+1$ are extracted.

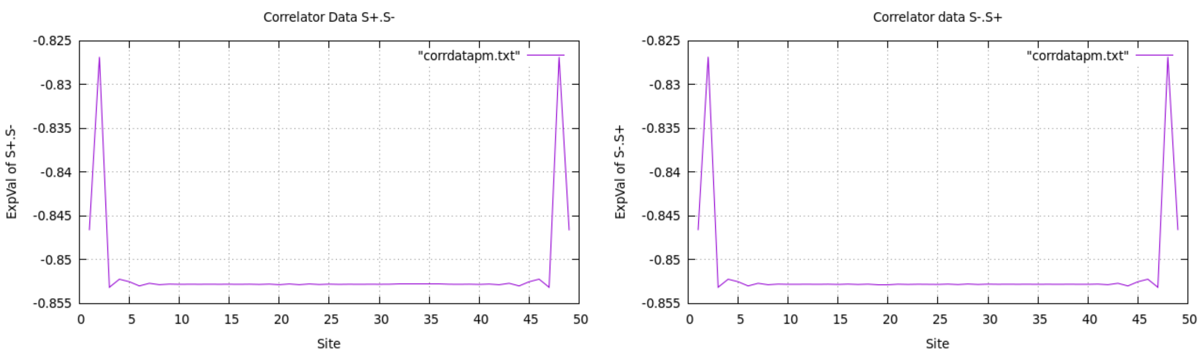

Figure 14: Correlation of S^+S^- and S^-S^+.

We see that both the plots given in Fig. 14 show the same pattern.

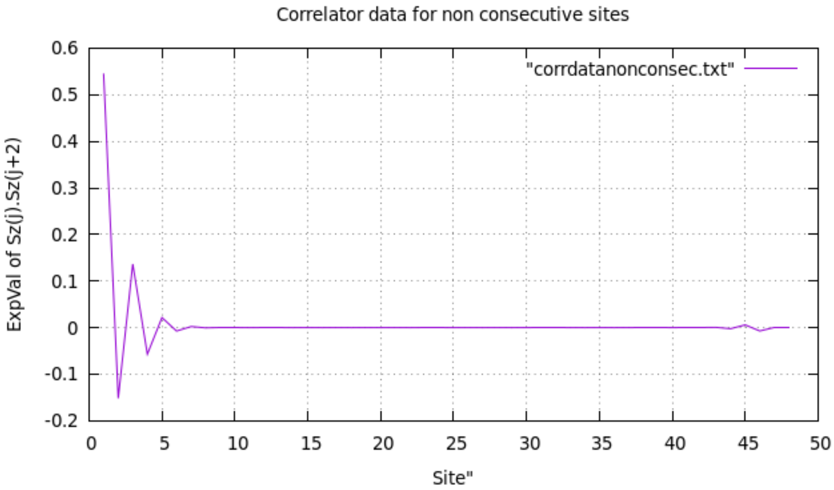

Figure 15: Spin-spin correlation for non-consecutive sites.

4.4 Spin-spin Correlation for Non-consecutive Sites

Since the model we considered is a one-dimensional one with local interactions, the spin-spin correlation (of S^z) for non-consecutive sites is 0, as expected. The only discrepancy is around the front boundary.

The explanation for this can be accounted by the fact that, while the correlation for the initial sites are being calculated, the quantum spin chain in its entirety hasn't fully formed. The calculation converges to zero as the spins start correlating with only their nearest neighbours, and steadily remains zero until the end. We show the result in Fig. 15.

4.5 The Ground State Energy

The ground state energy of the model is returned by the DMRG algorithm run, the details of which have been iterated previously. We show our results in Fig. 16. In Fig. 17 we show the ground state energy of the AKLT model per sweep of DMRG.

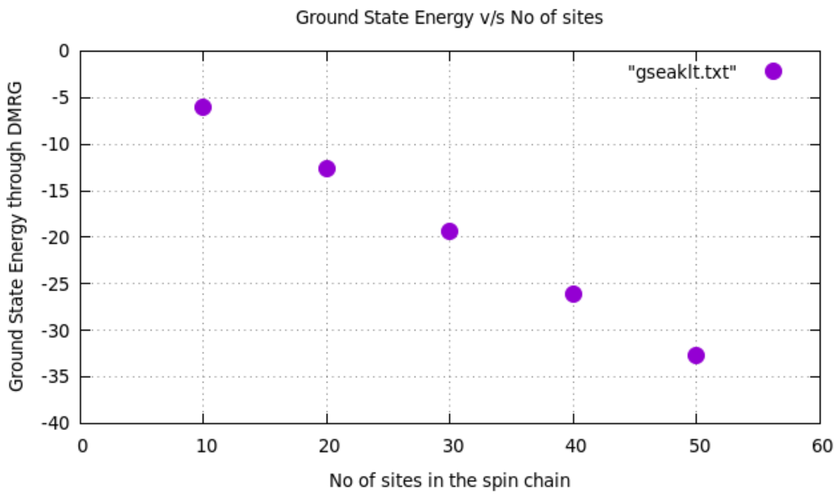

Figure 16: Ground state energy of the AKLT model for different number of sites.

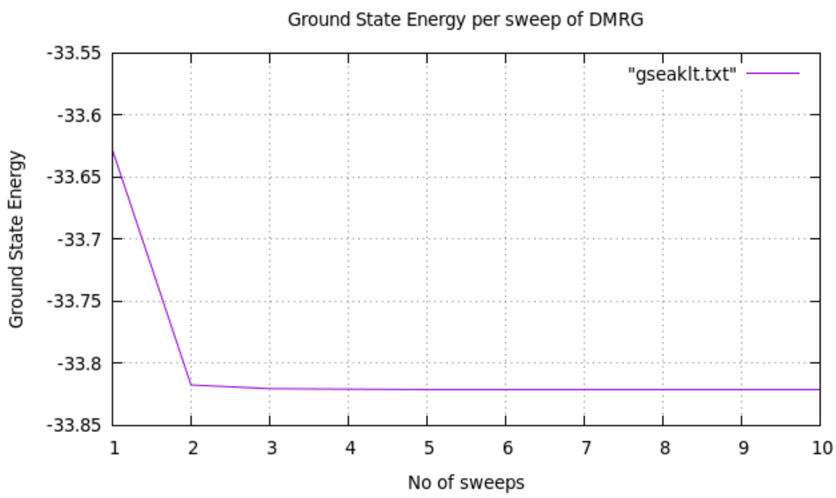

Figure 17: Ground state energy of the AKLT model per sweep of DMRG.

5 Addendum: The Majumder-Ghosh Model

The ground state of the Majumder-Ghosh model, like the AKLT one, is also an MPS. It is a frustrated spin chain defined by the Hamiltonian,[5]

$$H = \sum_i \left(\vec{S}^{[i]} \vec{S}^{[i+1]} + \frac{1}{2} \vec{S}^{[i]} \vec{S}^{[i+2]} \right), \tag{10}$$

where, $\vec{S}^{[i]}$ is the vector of spin-1/2 operators at site i.

5.1 Magnetization Per Site

The magnetization at each site, as computed for this model is shown in Fig. 18.

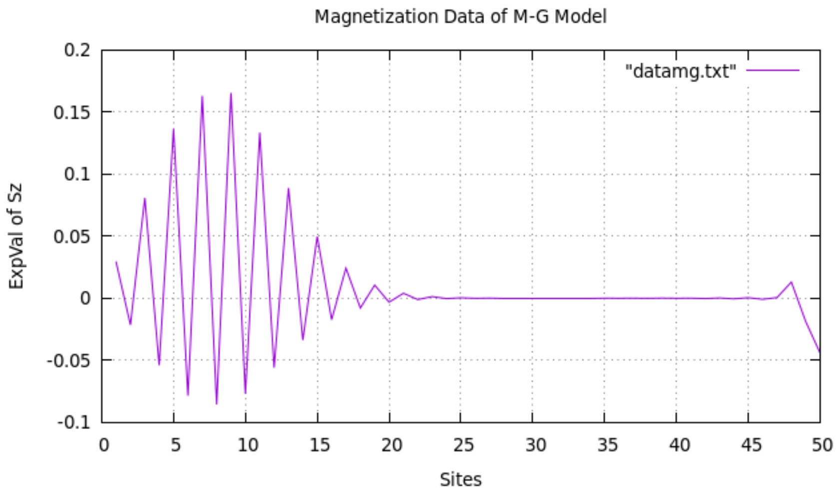

Figure 18: Magnetization per site for the Majumder-Ghosh model.

The Majumdar–Ghosh model, like the AKLT one, is notable because its ground states are exactly solvable; making both these models a useful starting point for understanding more complex spin models and phases.

6 Conclusion

Tensor network methods provide an efficient method of targeting the relevant corner in Hilbert spaces for quantum many-body systems. The implementation of these algorithms to probe low-energy states of one-dimensional models prove extremely useful in discerning their properties.

The properties of the ground state of the Affleck-Kennedy-Lieb-Tasaki (AKLT) model, as computationally discerned in this report, agree completely with the analytical derivations of them. The discrepancies noticed around the boundaries can be accounted by the fact that the analytical derivations consider closed boundary conditions, whereas the MPS constructed for this report is a random one, with open boundary conditions.

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